Reconciling Time's Passage with the Block Universe, Using Counterpart Theory

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1. The question of propositional temporalism

Propositional temporalism: Some propositions are temporarily true

$\exists p(p \wedge \text{sometimes } \neg p)$

Propositional eternalism: Every true proposition is always true

$\forall p(p \rightarrow \text{always } p)$

2. Is propositional temporalism obviously true?

You might think so, because of an argument like this:

> It is sunny in LA, although it isn't always sunny in LA
> So, some proposition—namely, the proposition that it is sunny in LA—is true but not always true

Propositional temporalists should regard this as sound. But propositional eternalists should treat it like the following unsound argument:

> He is rich, although not every male farmer is such that he is rich
> So, some proposition—namely, the proposition that he is rich—is true although not every male farmer is such that it is true.

3. Is the debate about propositional temporalism merely verbal?

A widespread reaction:

> Every true$_1$ proposition$_1$ is always true$_1$, but some true$_2$ propositions$_2$ are not always true$_2$.

But how can this work, since we surely need both of the following:

> For the proposition$_1$ that it's sunny in LA to be true$_1$ is for it to be sunny in LA
> For the proposition$_2$ that it's sunny in LA to be true$_2$ is for it to be sunny in LA

4. The Block Universe

Some important challenges to Propositional Temporalism come from certain hypotheses about "the fundamental level" which are arguably supported by discoveries in physics. I'll focus on one such hypothesis:

> *Pointillist Supervenience.* Every true proposition follows with metaphysical necessity from the Spacetime Truths, namely: (a) for each spacetime point, the truth that it exists and is a spacetime point; (b) for any two spacetime points, the truth

that they are distinct; (c) truths about geometric relations among spacetime points which can be summed up by assigning the set of points the structure of an differentiable manifold with an inextensible metric; (d) truths about some further relations among spacetime points which can be summed up by an assignment to the points of some physically distinguished tensor fields—e.g. the electromagnetic field and various "matter fields", and (e) a "that's all" truth (the proper characterization of which depends on further doctrines in metaphysics).

This is super-controversial, indeed probably false. But it'll do for today, since the kinds of modifications that look well motivated by considerations from physics don't seem to make things any more friendly to propositional temporalism.

The Supervenience Argument

> Any proposition that is necessitated by some eternal truths is eternal.
> <u>If Pointillist Supervenience is true, the Spacetime Truths are eternal.</u>
> So, if Pointillist Supervenience is true, every truth is eternal.

I'll be rejecting the second premise. (For a defence of the first, see Dorr and Goodman, 'Diamonds are Forever'). I'll address three challenges:

(i) What is it for a proposition to be going-to-be-false, or formerly-false, such that the Spacetime Truths could necessitate that some of them have that status?

(ii) *Which* of the Spacetime Truths are non-eternal, and which currently false propositions will be or have been true when the Spacetime Truths are not all true?

(iii) How can we think about the supervenience of propositions about people, tables, etc. on spacetime-theoretic propositions in such a way that familiar changes in these propositions are necessitated by the postulated changes in the distribution of truth-values over spacetime-theoretic propositions?

5. Temporal counterpart theory

> *Temporal counterpart theory:* for an object to [have had/be going to have] a qualitative property Q is for it to have a [past-/future-] counterpart with Q.

More generally:

> For objects $x_1 \ldots x_n$ to [have instantiated/be going to instantiate] a qualitative relation R is for there to be a [past-/future-] counterpairing f such that $f(x_1) \ldots f(x_n)$ stand in R.

and:

> For a proposition to [have been/be going to be] true is for it to be mapped to a truth by the "lift" of some [past-/future-] counterpairing.

- A hallmark of the view: necessarily, any true qualitative proposition is eternally true.

Given propositional counterpart theory, challenge (ii) reduces to the task of saying which functions from points to points are counterpairings, and challenge (i) reduces to the task of explaining why this follows from the Spacetime Truths.

6. A concern about arbitrariness, and a response invoking vagueness

Some possible spacetime structures would make this quite easy. For example, the geometry of "Newtonian spacetime" includes a natural partition of spacetime into 'rest-trajectories'. Here the obvious thing to say that the temporal counterpairings are the one-dimensional family of permutations that preserve geometric relations among spacetime points and map each point to a point on the same rest-trajectory.

But what if the geometrical and physical structure of spacetime doesn't provide any non-arbitrary way of singling out a one-dimensional group of permutations? Then we'd better say that 'is a [future-/past-] counterpairing' is *vague*. If so, the tense operators are also vague.

- Of course not just any permutation of spacetime points is a candidate to be in the extension of 'temporal counterpairing'. E.g.: in Minkowski spacetime, it's natural to think that each admissible interpretation of 'temporal counterpairing' picks out a group of *parallel time-like shifts*.
- We'd better make sure the vagueness of the tense-operators doesn't generate excessive vagueness in everyday discourse about the past and future. This may require positing "penumbral connections" which co-ordinate the vagueness in the tense operators with the vagueness of other words.

7. How facts about ordinary objects might supervene on spacetime facts

A proposed framework: each ordinary object is "associated" with some points making up a bounded 3d region. Familiar geometric properties of and relations among ordinary objects correspond straightforwardly to the spacetime-geometric relations among their associated regions: for example: x is *spatially inside* y iff x is associated with a nonempty region that is a subset of the one y is associated with.

- The set of points associated with *temporal counterparts* of an object form a 4d "worm". But the fact that a point is associated with a temporal counterpart of an object does not mean that the object will be or has been associated with the point!

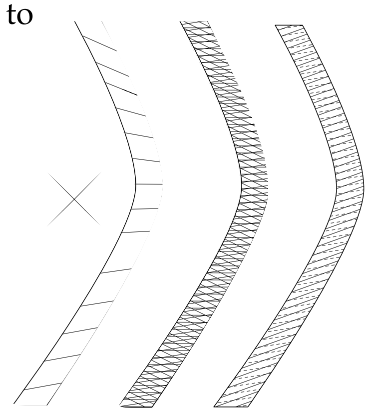
The *association profile* of an object is whatever property provides the necessary and sufficient conditions for a point to be associated with that object.

- Since ordinary objects can undergo qualitative change, their association profiles are *non-qualitative* properties of points.
- Also, since some ordinary objects are temporarily spatially within others, their association profiles must (typically) be *temporary* properties of points.

8. Relativity: two choice points.

First question: which slices through a given person-worm are associated with people?

> *Transverse slicing:* those approximately perpendicular to the trajectory of its centre of mass.
> *Criss-cross slicing:* all of them.
> *Parallel slicing:* those that belong to some family of parallel surfaces. (The question 'which family' is vague and context-sensitive.)



Second question: When a person is associated with a region R, what region is the n-second temporal counterpart of that person associated with?

> *Worm-distance counterpairings:* a subregion of the same worm that's (approximately) n-seconds ahead of R, as measured by the worm's proper time.
>
> *Slice-distance counterpairings:* a subregion of the same worm that is part of a simultaneity slice that's n seconds ahead of the simultaneity slice containing R.

(Note that *Transverse Slicing* is incompatible with *Slice-distance Counterpairings*.)

9. Against *Slice-Distance Counterpairings*

Consider Einstein's twins. At the biological age of 30, Castor's about to set out on a rocket-ship journey while Pollux stays at home.

Slice-Distance Counterpairings endorses both of the following natural speeches:

(1) In six years, Castor will be biologically 36, meeting someone who looks like him but is biologically 40.
(2) In ten years, Pollux will be biologically 40, meeting someone who looks like him but is biologically 36.

But if we take this seriously, we are forced to say that Castor and Pollux will never meet again. (Each will merely meet a temporal counterpart of the other.) In fact even the most trivial of journeys is enough to stop people from ever meeting again.

10. Some worries about *Criss-Cross Slicing*

(i) Hard to believe that 'How fast is my worm moving with respect to the slice I'm associated with' is simply an open (self-locating) question with a determinately right answer.

(ii) There's pressure to think that our credences about our true velocity should be invariant under boosts, in which case we have to be super-confident our "true velocity" is very high.

(iii) There's pressure to think that people with a high "true velocity" make lots of mistakes, e.g. about the shapes of things, or about how long it is since a star exploded.

11. A worry about *Parallel Slicing*

Considerations of charity motivate saying that there's *context-sensitivity* as well as vagueness. As used by a given person (or community of persons), the admissible resolutions of the vagueness of ordinary nouns like 'person' exclude objects associated regions relative to which that person's worm is moving too fast.

Upshot: 'Person' as used by communities of people who are moving fast relative to us doesn't express *being a person* (on any admissible precisification). Likewise, members of these community don't refer to themselves (on any admissible precisification) when they say 'I'.

Although this seems weird, I think it's something physicalists have independent reason to get used to.